Qualcomm Technologies, Inc.

Unified Sensor Test Application (USTA)

User Guide

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Revision history

Revision	Date	Description
A	February 2018	Initial release
B	September 2020	Numerous changes have been made to this document revision. It should be read in its entirety.

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1 Introduction

1.1 Purpose

This document describes how to use the Unified Sensor Test Application (USTA) for sensor streaming. USTA is part of current software releases and is enabled with default Qualcomm® builds provided for Android.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Shading indicates content that has been added or changed in this revision of the document.

1.3 Technical assistance

For assistance or clarification on information in this document, submit a case to Qualcomm Technologies, Inc. (QTI) at https://createpoint.qti.qualcomm.com/.

If you do not have access to the CDMATech Support website, register for access or send email to support.cdmatech@qti.qualcomm.com.

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2 Getting started

USTA dynamically supports all sensor types available from the Sensors Execution Environment (SEE) client API. USTA creates and sends requests to any sensor available with the SEE framework, parses all the events back to the client, and displays the events in the application UI and adb logcat.

The USTA source code is available in the following Android build paths:

- Android R

  `LINUX\android\vendor\qcom\proprietary\sensors-see\USTA`

- Android Q

  `LINUX\android\vendor\qcom\proprietary\commonsys\sensors-see\USTA`
  `LINUX\android\vendor\qcom\proprietary\commonsys-intf\sensors-see\USTA`

To start USTA, open the USTA application on the device.

2.1 USTA tabs

The following describes the tabs on the application UI.

- USTA-TAB − Lists all the sensors that are exposed at the client level and identified by USTA. See Chapter 3 for details.

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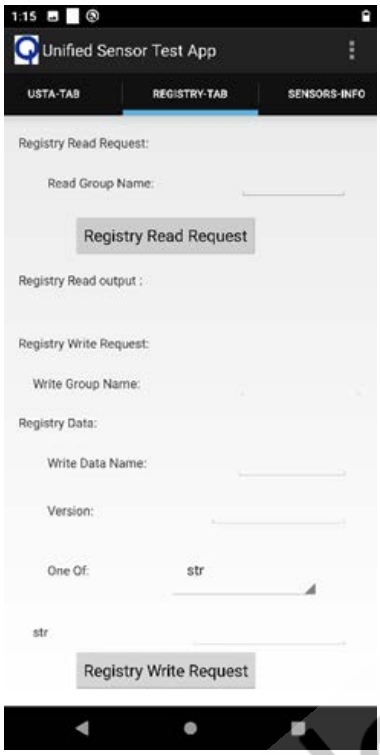
- REGISTRY-TAB − Allows for reading and/or writing to the registry based on the registry group name. See Chapter 6 for details.



- SENSORS-INFO − Displays the attributes for sensors that are identified by USTA. See Chapter 7 for details.

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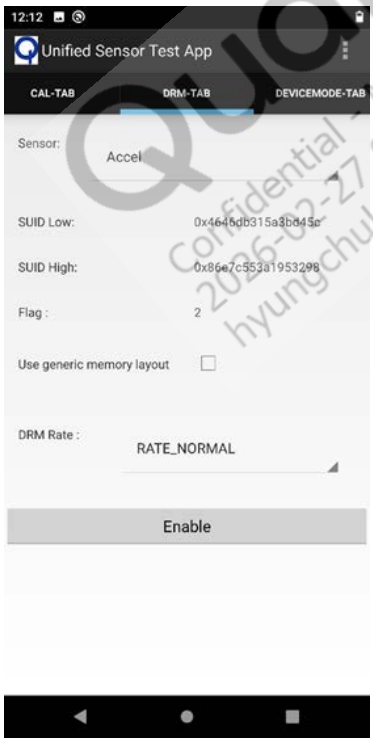
- CAL-TAB − Executes the user calibration for accelerometer via JNI.

- DRM-TAB − Establishes the direct channel (LLCM) connection. See Chapter 8 for details.



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- DEVICEMODE-TAB − Sets the device mode and state sensor used by the magnetometer sensor driver.

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3 Using USTA UI to stream sensors

1. Start USTA. The default USTA-TAB tab shows the sensors listed on USTA.

The sensor name is concatenated with its vendor name. For all sensors, the name appears in front of Sensor field in a sensor_name-vendor_name format. The On-Change, HW_ID, and Rigid-Body-Type parameters for the selected sensor are also shown below the sensor field. The SUID for a selected sensor appears in the SUID Low and SUID High text fields. Other options are available as either drop-down lists or check boxes, or as fields in which information is entered, based on each sensor's .proto file.

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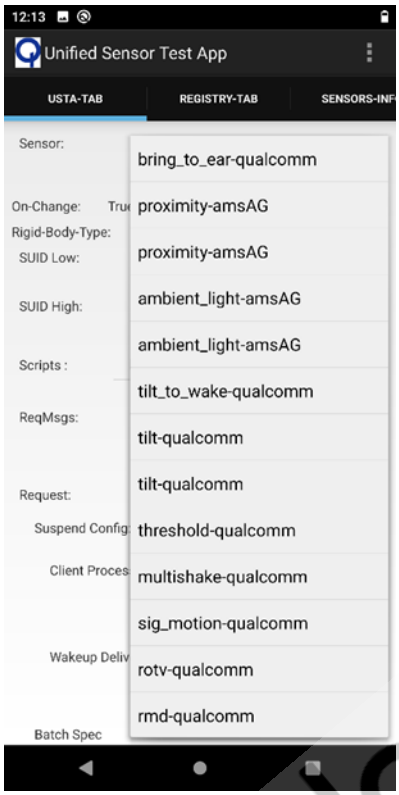
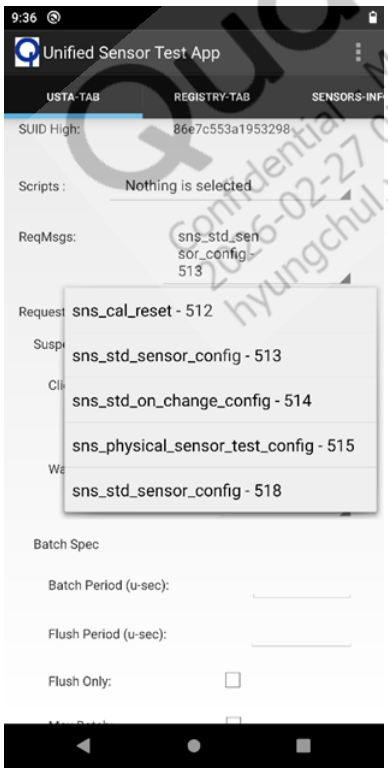
2. To stream a sensor using the USTA UI, select one of the sensors in the sensor list.



Multiple message requests appear in ReqMsgs.



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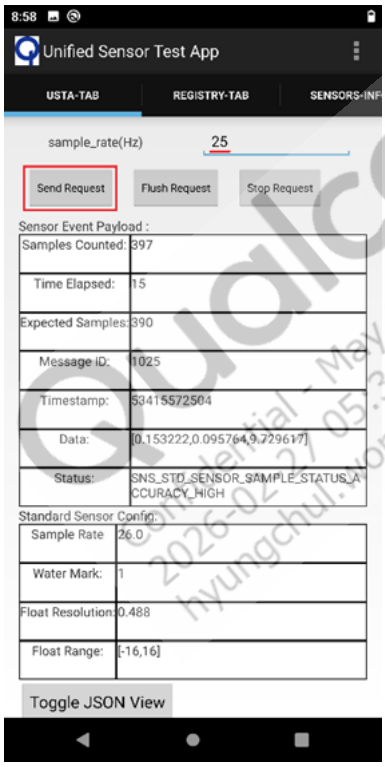
a. To view the payload fields, select one of the request messages in ReqMsgs, for example, **sns_std_sensor_config-513**.

   Payload fields vary according to the per-request message based on the .proto files. For most sensors, sns_std_sensor_config-513, sns_std_on_change_request-514, and sns_physical_sensor_test_config-515 are used for streaming. Refer to the related sensor .proto file before making a request.

b. In Request > Suspend Config, the Client Processor and Wakeup Delivery type must be selected. For streaming an accelerometer sensor to the application processor, select **sns_std_client_processor_apss** and then select **sns_client_delivery_wakeup** to wake up the application processor when the samples arrive.

c. For accelerometer, fields in Batch Spec are optional. When requesting a batching request, complete these fields. For example, if data is needed every 1 sec, set the batch period to 1000000 (microseconds).

d. In Payload, type the sampling rate and click **Send Request**.



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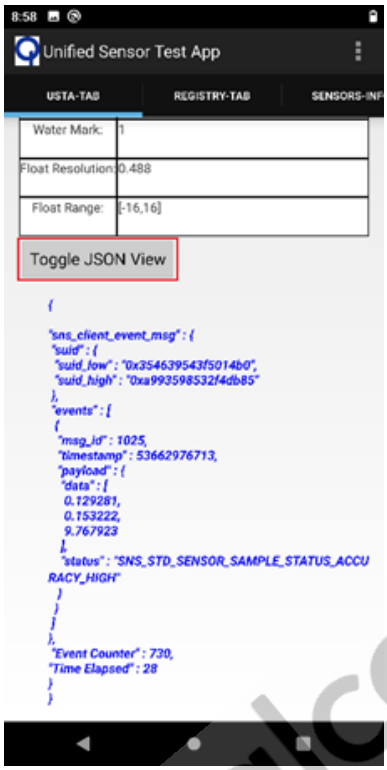
A sample_rate request for 25 Hz is sent to the driver. Events from the accelerometer sensor streaming on the USTA UI will appear. The JSON view can be toggled by pressing the **Toggle JSON View**.



To view the request and events in adb logcat, issue an `adb logcat -c && adb logcat -v time` command.

Request

```
D USTA_Native_Debug:  encoded input message for sns_std_sensor_config
is generated
D USTA_Native_Debug: payload is not empty
D USTA_Native_Debug:  encoded input message for
sns_client_request_msg is generated
D USTA_Native_Debug:
D USTA_Native_Debug: "sns_client_request_msg" : {
D USTA_Native_Debug:  "suid" : {
D USTA_Native_Debug:   "suid_low" : "0x4646db315a3bd45c",
D USTA_Native_Debug:   "suid_high" : "0x86e7c553a1953298"
D USTA_Native_Debug:  },
D USTA_Native_Debug:  "msg_id" : 513,
D USTA_Native_Debug:  "susp_config" : {
D USTA_Native_Debug:   "client_proc_type" :
"SNS_STD_CLIENT_PROCESSOR_APSS",
D USTA_Native_Debug:   "delivery_type" : "SNS_CLIENT_DELIVERY_WAKEUP"
D USTA_Native_Debug:  },
D USTA_Native_Debug:  "request" : {
```

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```
D USTA_Native_Debug:   "payload" : {
D USTA_Native_Debug:    "sample_rate" : 25.000000
D USTA_Native_Debug:   }
D USTA_Native_Debug:  }
D USTA_Native_Debug: }
```

Sensor events

```
D USTA_Native_Debug: accel-STMicro event message count (image
(05_24)) is 10
D USTA_Native_Debug: log_message_size 342
D USTA_Native_Debug:
D USTA_Native_Debug: "sns_client_event_msg" : {
D USTA_Native_Debug:  "suid" : {
D USTA_Native_Debug:   "suid_low" : "0x4646db315a3bd45c",
D USTA_Native_Debug:   "suid_high" : "0x86e7c553a1953298"
D USTA_Native_Debug:  },
D USTA_Native_Debug:  "events" : [
D USTA_Native_Debug:   {
D USTA_Native_Debug:    "msg_id" : 1025,
D USTA_Native_Debug:    "timestamp" : 15438466252,
D USTA_Native_Debug:    "payload" : {
D USTA_Native_Debug:     "data" : [
D USTA_Native_Debug:      0.268139,
D USTA_Native_Debug:      0.143646,
D USTA_Native_Debug:      9.643430
D USTA_Native_Debug:     ],
D USTA_Native_Debug:     "status" :
"SNS_STD_SENSOR_SAMPLE_STATUS_ACCURACY_HIGH"
D USTA_Native_Debug:    }
D USTA_Native_Debug:   }
D USTA_Native_Debug:  ]
D USTA_Native_Debug: }
```

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3. To stop streaming, click **Stop Request**.

To view the events in adb logcat, issue a `stopRequest` command.

```
I USTA_Native_Info:  stopRequest Start
I USTA_Native_Info: stopRequest UI Mode it is
I USTA_Native_Info:  stopRequest End
```

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3.1 Suspend configuration fields

The Client Processor type and Wakeup Delivery type are mandatory fields and must be selected.

3.1.1 Client Processor type

The Client Processor type is the processor on which the client resides.

1. In Client Processor, click the down arrow to view the list of supported clients.

2. Generally, the test application resides on an application processor; therefore, select **sns_std_client_processor_apss** as the client processor type.

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3.1.2 Wakeup Delivery type

The SEE framework uses the Wakeup Delivery type to determine whether to send events to the client processor in Suspend mode.

1. In Wakeup Delivery, click the down arrow to view the list of wakeup and non-wakeup delivery types.

2. Select the appropriate Wakeup Delivery type.
   - sns_client_delivery_wakeup – Used if the events should be sent to the client whenever events are available. Generally, events are available at the sampling rate.
     - If a batch_period is larger than the system capacity is requested, all data is sent upon capacity exhaustion.
     - The flush_period is effectively ignored, as unsent batched data does not accrue in the buffer.
   - sns_client_delivery_no_wakeup – Used if events should be sent to the client only if the client processor is awake; otherwise, batching occurs.

When the client processor exits Suspend mode, all the events are flushed out.

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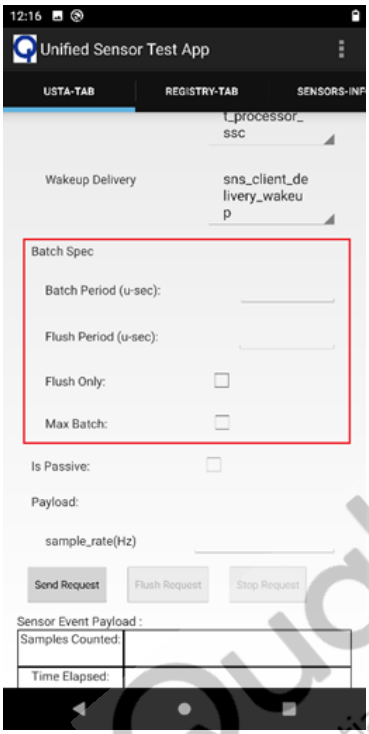
3.2 Batch specification fields

The Batch Period, Flush Period, Flush Only, and Max Batch fields are part of the batch_spec message. The batch_spec message is an optional field. Therefore, it is acceptable to leave these fields blank. If the client is interested in batched data, these fields are required.

If the Flush Period, Flush Only, and Max Batch fields are provided in the batch_spec message, the Batch Period is a required field. Otherwise, the sensor does not stream.



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Batch Period

The Batch Period is an unsigned integer value that is used to set the batching interval. Events are delivered to the client according to this batching interval. Batch Period 0 indicates that no batching occurs.

Flush Period

The default Flush Period is set to the Batch Period value. This parameter tells the sensor to always provide the latest flush period worth of data. If the Flush Period is less than the Batch Period, the sensor may not publish any events to clients, depending on other concurrent clients.

Flush Only

If the Flush Only field is set to True, the sensor sends data to the client only on receiving a flush request or if the sensor cannot accumulate a flush period worth of data. The default value for this field is False.

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Max Batch

If the Max Batch field is set to True, the sensor operates at maximum batching capacity for all requests. The absence of this field is equivalent to max_batch = False. If a request has max_batch = True and flush_only = True, flush_only takes precedence.

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4 Using USTA command line to stream sensors

USTA provides a command-line interface. The following list of high-level command-line instructions are currently available.

- To start USTA, type the following command:

  ```
  adb root
  adb shell am start-foreground-service
  com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
  ```

- To list all the sensors available on USTA, type the following command:

  ```
  adb shell am broadcast -a
  com.qualcomm.qti.usta.core.intent.GETSENSORLIST
  ```

- To get attribute information for a specific sensor, type the following command:

  ```
  adb shell am broadcast -a
  com.qualcomm.qti.usta.core.intent.GETSENSORATTRIBUTES
  ```

  Select a sensor by providing the sensor handle or SUID information (both obtained from `GETSENSORLIST`). Either argument is required, that is, sensor handle (integer value with `sensorhandle` as the string name) or SUID (two string arguments with `suid_low` and `suid_high` as the string name).

  Generalized instructions:

  ```
  --ei 'sensorhandle' <sensor_handle_number>
  --es 'suid_low' <suid_low_number> --es 'suid_high' <suid_high_number>
  ```

- To close USTA, type the following command:

  ```
  adb shell am force-stop com.qualcomm.qti.usta
  ```

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4.1 Handle number

Identify the handle number for a sensor by running a `GETSENSORLIST` command. Use the handle number to stream the sensor.

The following sequence provides a detailed example of how to stream accelerometer from a USTA command line.

1. Open two command windows. In one command window, type `adb logcat -c && adb logcat-v time` and in the other window, type the following commands in sequence. (**Note:** The order of the command-line instructions is important.)
2. To start a USTA command-line service, type the following command:

   ```
   adb shell am start-foreground-service
   com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
   ```

   Expected log from the adb command:

   ```
   Starting service: Intent { act=android.intent.action.MAIN
   cat=[android.intent.category.LAUNCHER]
   cmp=com.qualcomm.qti.usta/.core.USTACmdLineService }
   ```

3. To get the sensors list, type the following command:

   ```
   adb shell am broadcast -a
   com.qualcomm.qti.usta.core.intent.GETSENSORLIST
   ```

   Expected log from the adb command:

   ```
   Broadcasting: Intent
   { act=com.qualcomm.qti.usta.core.intent.GETSENSORLIST flg=0x400000 }
   Broadcast completed: result=0
   ```

   In an adb logcat window, all the sensor types with handle numbers appear.

   ```
   ...
   D/USTA_APP_DEBUG( 5350): Sensor Handle:21 Sensor Name:motion_detect-
   STMicro Sensor SUID Low:f84fdb82cd8c500a Sensor SUID
   High:2838874eefafb0b4
   D/USTA_APP_DEBUG( 5350): Sensor Handle:22 Sensor Name:gyro-STMicro
   Sensor SUID Low:9d40d025dbe2b188 Sensor SUID High:3e5ef860b79f9ba
   D/USTA_APP_DEBUG( 5350): Sensor Handle:23 Sensor Name:accel-STMicro
   Sensor SUID Low:4646db315a3bd45c Sensor SUID High:86e7c553a1953298
   ...
   ```

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4. To get the sensor attributes for the selected sensor handle, type the following command:

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GETSENSORATTRIBUTES --ei
'sensorhandle' 23
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.GETSENSORATTRIBUTES flg=0x400000
(has extras) }
Broadcast completed: result=0
```

In an adb logcat window, all the attributes for the selected sensor appear.

5. To get the request message for the selected sensor handle, type the following command:

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GETREQUESTMSGS --ei 'sensorhandle' 23
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.GETREQUESTMSGS flg=0x400000
(has extras) }
Broadcast completed: result=0
```

In an adb logcat window, all the supported request parameters for the selected sensor appear.

6. To enable the selected sensor, type the following command:

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --ei
'sensorhandle' 23 --ei 'msg_id' 513 --ei 'client' 0 --ei 'wakeup' 0 --ei
'batch_period' 0   --ei 'flush_period' 0 --ef 'sample_rate' 2.0 --es
'logfile' 'sensorLog.json'
```

Expected log from the adb command:

```
Broadcasting: Intent { act=com.qualcomm.qti.usta.core.intent.ENABLE
flg=0x400000 (has extras) }
Broadcast completed: result=0
```

The key parameters are sensorhandle 23, msg_id 513 (for sns_std_sensor_config), similar to the USTA UI options.

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In an adb logcat window, the accelerometer events streaming appears.

```
D/USTA_Native_Debug( 2480): log_message_size 342
D/USTA_Native_Debug( 2480):
D/USTA_Native_Debug( 2480): "sns_client_event_msg" : {
D/USTA_Native_Debug( 2480):  "suid" : {
D/USTA_Native_Debug( 2480):   "suid_low" : "0x4646db315a3bd45c",
D/USTA_Native_Debug( 2480):   "suid_high" : "0x86e7c553a1953298"
D/USTA_Native_Debug( 2480):  },
D/USTA_Native_Debug( 2480):  "events" : [
D/USTA_Native_Debug( 2480):   {
D/USTA_Native_Debug( 2480):    "msg_id" : 1025,
D/USTA_Native_Debug( 2480):    "timestamp" : 38868009771,
D/USTA_Native_Debug( 2480):    "payload" : {
D/USTA_Native_Debug( 2480):     "data" : [
D/USTA_Native_Debug( 2480):      0.450091,
D/USTA_Native_Debug( 2480):      0.244198,
D/USTA_Native_Debug( 2480):      9.633853
D/USTA_Native_Debug( 2480):     ],
D/USTA_Native_Debug( 2480):     "status" :
"SNS_STD_SENSOR_SAMPLE_STATUS_ACCURACY_HIGH"
D/USTA_Native_Debug( 2480):    }
D/USTA_Native_Debug( 2480):   }
D/USTA_Native_Debug( 2480):  ]
D/USTA_Native_Debug( 2480): }
D/USTA_Native_Debug( 2480): accel-STMicro event message count (image
(05_24)) is 162
```

7. To disable the selected sensor, type the following command:

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.DISABLE --ei
'sensorhandle' 23
```

Expected log from the adb command:

```
Broadcasting: Intent { act=com.qualcomm.qti.usta.core.intent.DISABLE
flg=0x400000 (has extras) }
Broadcast completed: result=0
```

In an adb logcat window, logs shown indicate a disabling request for `sensorhandle` 23.

```
I/USTA_APP_INFO( 6002): DISABLE intent
D/USTA_APP_DEBUG( 6002): stop request from command line for sensor 23
I/USTA_Native_Info( 2480):  stopRequest Start
I/USTA_Native_Info( 2480): stopRequest UI Mode it is
D/USTA_Native_Debug( 2480): is_qmi_disable_api_enabled is false
I/USTA_Native_Info( 2480):  stopRequest End
D/USTA_APP_DEBUG( 6002): after stop request for the sensorhandle 23
```

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8. To stop USTA command-line service, type the following command:

```
adb shell am stopservice
com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
```

Expected log from the adb command:

```
Stopping service: Intent { act=android.intent.action.MAIN
cat=[android.intent.category.LAUNCHER]
cmp=com.qualcomm.qti.usta/.core.USTACmdLineService }
Service stopped
```

4.2 SUID number

Identify the SUID for a sensor by running a `GETSENSORLIST` command. Use the SUID to stream the sensor.

```
adb shell am start-foreground-service
com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.GETSENSORLIST
```

In an adb logcat window, all sensor types listed with SUIDs for accelerator appear along with the handle number.

```
...
D/USTA_APP_DEBUG( 6002): Sensor Handle:21 Sensor Name:motion_detect-STMicro
Sensor SUID Low:f84fdb82cd8c500a Sensor SUID High:2838874eefafb0b4
D/USTA_APP_DEBUG( 6002): Sensor Handle:22 Sensor Name:gyro-STMicro Sensor
SUID Low:9d40d025dbe2b188 Sensor SUID High:3e5ef860b79f9ba
D/USTA_APP_DEBUG( 6002): Sensor Handle:23 Sensor Name:accel-STMicro Sensor
SUID Low:4646db315a3bd45c Sensor SUID High:86e7c553a1953298
...
```

The following example shows a request made with a SUID (low and high) for accelerometer:

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GETSENSORATTRIBUTES --es 'suid_low'
4646db315a3bd45c --es 'suid_high' 86e7c553a1953298
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.GETREQUESTMSGS
--es 'suid_low' 4646db315a3bd45c --es 'suid_high' 86e7c553a1953298
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --es
'suid_low' 4646db315a3bd45c --es 'suid_high' 86e7c553a1953298 --ei 'msg_id'
513 --ei 'client' 0 --ei 'wakeup' 0 --ei 'batch_period' 0 --ei
'flush_period' 0 --ef 'sample_rate' 26.0 --es 'logfile' 'sensorLog.json'
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.DISABLE --es
'suid_low' 4646db315a3bd45c --es 'suid_high' 86e7c553a1953298
adb shell am stopservice
com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
```

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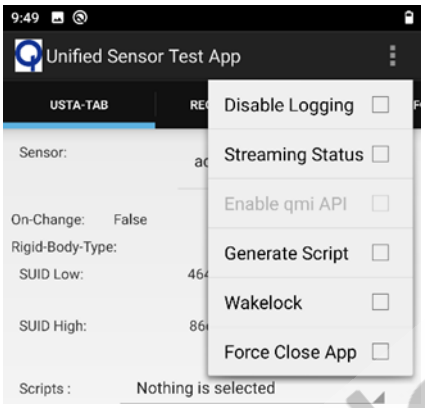
5 Other options and menu settings

USTA supports Disable Logging, Streaming Status, Generate Script, Wakelock, and Force Close App. These menu items can be selected by clicking the ellipsis in the upper right corner of the UI.



5.1 Disable Logging

Disable Logging controls USTA logging, such as adb logcat logs and logging into /mnt/sdcard/.

From the UI

To disable logs from the UI, select **Disable Logging** before clicking the USTA-TAB. Change the logging status at any time during the execution of USTA by selecting or clearing **Disable Logging**.

From the command line

- To disable logging from a command line, type the following command before running the start service command:

  ```
  adb shell setprop sensors.log.disable 1
  ```

- To disable logging during the execution of USTA from a command line, type the following command-line instruction. Switch between enable and disable logging at any time.
  - To disable logging, type the following:

    ```
    adb shell am broadcast -a
    com.qualcomm.qti.usta.core.intent.DISABLELOGGING --ei
    ```

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```
'disable_logging'
1
```

- To enable logging, type the following:

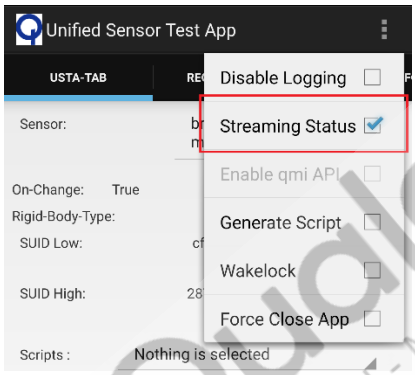
```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.DISABLELOGGING --ei
'disable_logging'
0
```

In both the cases, logging is enabled by default.

Using a command line is recommended when USTA is used for power measurement purposes.

5.2 Streaming Status

Streaming Status controls whether sensor event messages appear on the UI. By default, events appear on USTA. To disable event messages on the USTA UI, clear the **Streaming Status** checkbox.



5.3 Generate Script

To generate scripts for automation test purposes:

1. Select **Generate Script**.
2. On USTA-TAB, select any sensor, fill out all required fields, and click **Send Request**.
3. When prompted, enter the script name and click **Save**. The script will be saved at /data/sensors/scripts.

To load the script:

1. Select the sensor from sensors list on the USTA-TAB.
2. Select the saved script from the Script drop-down list. Once selected, the required UI fields will be automatically populated from the selected script.
3. Click **Send Request**.
4. When prompted, enter a script name. Click **No** if you do not want to create another script and the request will be sent.

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5.4 Wakelock

Select this menu item to prevent the APSS from going into suspend.

5.5 Force Close App

Force Close App closes USTA and removes the application from the recent applications window.

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6 Registry functionality

USTA supports reading and writing/modifying the registry based on the registry group name.

From the UI

- Registry Read

  Type a group name in the **Read Group Name** field in which you want to read, and click **Registry Read Request**. Registry details appear on the USTA UI screen.

  For example, for accelerometer (LSM6DST), **lsm6dst_0.accel** in the group name field shows the event message on the UI and in adb logcat.

- Registry Write

  Type a group name in **Write Group Name** in which you want to write an item. Type the data item name in **Write Data Name** and version number in **Version**. (The version number must be one greater than the previous one.). After all the fields are complete, click **Registry Write request**.

  To verify whether the registry write for this data item is successful, send a **Registry Read** for the group name and cross-check the values entered.

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From the command line

Registry groups can be read from a command line. Similar commands must be used for a command line, with the exception that the sensor_handle is used for a registry sensor.

Generic commands

- Read

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --ei
'sensorhandle' <registry_sensor_Handle_number> --ei 'msg_id' 512 --ei
'client' 1 --ei 'wakeup' 0 --es 'name' <registry_group_name>
```

- Write

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --ei
'sensorhandle' <registry_sensor_Handle_number> --ei 'msg_id' 513 --ei
'client' 1 --ei 'wakeup' 0 --es 'sub_message_name' <group_name> --es
'data_name' <data_item_name> --ei 'version' <version_number> --es
'oneof_name' <one_of_filed_string> --es 'one_of_value' <one_of_value>
```

For example, the following command is used for lsm6dst_0.accel.config:

- Read

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --ei
'sensorhandle' 15 --ei 'msg_id' 512 --ei 'client' 1 --ei 'wakeup' 0 --es
'name' lsm6dst_0.accel.config     // Registry Read
```

a. To start the USTA command-line service, type the following:

```
adb shell am start-foreground-service
com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
```

b. To get the sensor list and find the registry sensor, type the following:

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GETSENSORLIST
```

```
...
D/USTA_APP_DEBUG( 6592): Sensor Handle:15 Sensor Name:registry-
qualcomm Sensor SUID Low:5e2541b4701e2275 Sensor SUID
High:e12754a7007f2759
...
```

c. To read a registry group using a registry sensor, for example, lsm6dst_0.accel.config, type the following:

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --
ei 'sensorhandle' 15 --ei 'msg_id' 512 --ei 'client' 1 --ei 'wakeup'
0 --es 'name' lsm6dst_0.accel.config
```

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An adb logcat window shows the registry output.

```
D/USTA_Native_Debug( 2466): "sns_client_event_msg" : {
D/USTA_Native_Debug( 2466):  "suid" : {
D/USTA_Native_Debug( 2466):   "suid_low" : "0x5e2541b4701e2275",
D/USTA_Native_Debug( 2466):   "suid_high" : "0xe12754a7007f2759"
D/USTA_Native_Debug( 2466):  },
D/USTA_Native_Debug( 2466):  "events" : [
D/USTA_Native_Debug( 2466):   {
D/USTA_Native_Debug( 2466):    "msg_id" : 514,
D/USTA_Native_Debug( 2466):    "timestamp" : 4029878619,
D/USTA_Native_Debug( 2466):    "payload" : {
D/USTA_Native_Debug( 2466):     "name" : "lsm6dst_0.accel.config",
D/USTA_Native_Debug( 2466):     "data" : {
D/USTA_Native_Debug( 2466):      "items" : [
D/USTA_Native_Debug( 2466):       {
D/USTA_Native_Debug( 2466):        "name" : "is_dri",
D/USTA_Native_Debug( 2466):        "version" : 0,
D/USTA_Native_Debug( 2466):        "sint" : 1
```

d. To disable the registry sensor, type the following:

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.DISABLE --ei 'sensorhandle' 15
```

e. To stop the USTA command-line service, type the following:

```
adb shell am stopservice com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
```

- Write

To write to a registry group using a registry sensor, for example, lsm6dst_0.accel.config, type the following:

```
adb shell am broadcast -a com.qualcomm.qti.usta.core.intent.ENABLE --ei 'sensorhandle' 15 --ei 'msg_id' 513 --ei 'client' 1 --ei 'wakeup' 0 --es 'sub_message_name' lsm6dst_0.accel.config --es 'data_name' is_dri --ei 'version' 4 --es 'oneof_name' sint --es 'one_of_value' 20
```

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7 Sensor attributes

The SENSORS-INFO tab shows the attributes for each sensor identified by the USTA UI. By default, a sensor with a zero (0) handle displays its attributes. To view the attributes of a specific sensor, select a sensor from the drop-down list. The following examples show attributes for a proximity sensor and accelerometer sensor.

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8 Direct report channel (DRC)

DRM-TAB in USTA can be used to establish a direct channel (LLCM) connection.

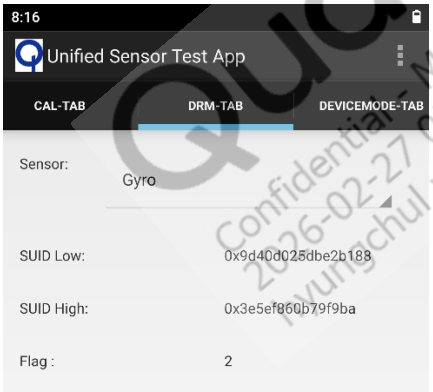
From the UI

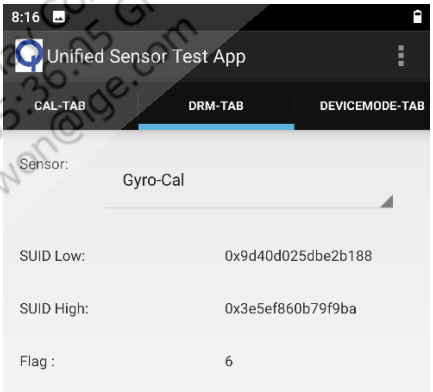
1. Select a sensor from the drop-down menu.
2. Select the rate at which you would like to stream the sensor.
3. Click **Enable** to start streaming.
4. Click **Disable** to stop streaming.

See Section 8.1 for QXDM Professional™ LLCM logs.

Direct channel supports accelerometer, gyroscope, and magnetometer. To view the SUID for any direct channel-supported sensor, select the sensor from the sensor drop-down list in the DRM-TAB.

Note: Gyro and Gyro-Cal use the same SUID but the Flag field on the DRM-TAB will change automatically to show 2 (non-calibration sensor) and 6 (calibration sensor), respectively.





From the command line

For direct channel command line instructions, 'channel_id' can be set from 1 to 32 and mapped to any sensor. Also, ‘flags’ can be 0x2 for non-calibration sensors and 0x6 for calibration sensors

Direct channel rate can be one of the following:

- RATE_STOP − 0
- RATE_NORMAL − 1
- RATE_FAST − 2

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- RATE_VERY_FAST − 3

Example

1. Start a USTA command-line service.

```
adb shell am start-foreground-service
com.qualcomm.qti.usta/com.qualcomm.qti.usta.core.USTACmdLineService
```

Expected log from the adb command:

```
Starting service: Intent { act=android.intent.action.MAIN
cat=[android.intent.category.LAUNCHER]
cmp=com.qualcomm.qti.usta/.core.USTACmdLineService }
```

2. Open the sensors list and note the SUID of a direct channel supported sensor.

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GETSENSORLIST
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.GETSENSORLIST flg=0x400000 }
Broadcast completed: result=0
```

In an adb logcat window, the following will appear:

```
...
D/USTA_APP_DEBUG( 5357): Sensor Handle:23 Sensor Name:accel-STMicro
Sensor SUID Low:4646db315a3bd45c Sensor SUID High:86e7c553a1953298
...
```

3. Enable the sensor.

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.GET_DRMATTRIBUTES --es 'suid_low'
4646db315a3bd45c --es 'suid_high' 86e7c553a1953298
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.GET_DRMATTRIBUTES flg=0x400000
(has extras) }
Broadcast completed: result=0
```

In an adb logcat window, the following will appear:

```
I/USTA_APP_INFO( 5357): GET_DRMATTRIBUTES intent
I/USTA_APP_INFO( 5357): getDrmSupportedRate called
I/USTA_APP_INFO( 5357): sensorSuid low 5063975836038452316 & sensorSuid
high -8725788790023179624
I/USTA_APP_INFO( 5357): supported max rate 0
I/USTA_APP_INFO( 5357): getDrmSupportedRate done
```

4. Create direct channel.

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.CREATE_DRMCHANNEL --ei 'buffersize'
0x1000 --ei 'channel_id' 1
```

Expected log from the adb command:

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```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.CREATE_DRMCHANNEL flg=0x400000
(has extras) }
Broadcast completed: result=0
```

In an adb logcat window, the following will appear:

```
I/USTA_APP_INFO( 5357): CREATE_DRMCHANNEL intent
I/USTA_APP_INFO( 5357): createDrmChannel called
I/USTA_APP_INFO( 5357): buffer size 4096
I/USTA_APP_INFO( 5357): channel_id 1
I/USTA_APP_INFO( 5357): get drm channel handle for channel_id 1
I/USTA_APP_INFO( 5357): SensorLowLatencyJNI - getchannel is  -1
```

5. Configure direct channel.

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.CONFIGURE_DRMCHANNEL --ei 'channel_id'
1 --ei 'drmrate' 3 --ei 'flags' 0x2 --es 'suid_low' 4646db315a3bd45c --
es 'suid_high' 86e7c553a1953298 --ei 'channel_type' 0 --ei
'sensor_handle' 0x1
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.CONFIGURE_DRMCHANNEL
flg=0x400000 (has extras) }
Broadcast completed: result=0
```

In an adb logcat window, the following will appear:

```
I/USTA_APP_INFO( 5357): CONFIGURE_DRMCHANNEL intent
I/USTA_APP_INFO( 5357): configureDrmChannel called
I/USTA_APP_INFO( 5357): channel_id 1, samplerate 3, flags 2
I/USTA_APP_INFO( 5357): String suid_low 4646db315a3bd45c & suid_high
86e7c553a1953298
I/USTA_APP_INFO( 5357): sensorSuid low 5063975836038452316 & sensorSuid
high -8725788790023179624
I/USTA_APP_INFO( 5357): lowLatChannelSampleRate samplerate selected 1250
I/USTA_APP_INFO( 5357): configureDrmChannel completed
```

6. Delete direct channel.

```
adb shell am broadcast -a
com.qualcomm.qti.usta.core.intent.DELETE_DRMCHANNEL --ei 'channel_id' 1
```

Expected log from the adb command:

```
Broadcasting: Intent
{ act=com.qualcomm.qti.usta.core.intent.DELETE_DRMCHANNEL flg=0x400000
(has extras) }
Broadcast completed: result=0
```

In an adb logcat window, the following will appear:

```
I/USTA_APP_INFO( 5357): DELETE_DRMCHANNEL intent
I/USTA_APP_INFO( 5357): deleteDrmChannel called
```


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```
I/USTA_APP_INFO( 5357): channel_id 1
I/USTA_APP_INFO( 5357): closing drm channel -1
I/USTA_APP_INFO( 5357): IMPROPERINPUT: drm channel not availalbe for 1
I/USTA_APP_INFO( 5357): deleteDrmChannel done
```

8.1 Sample QXDM Pro log

The following are sample QXDM Pro logs for a gyro request.

- Starting stream

```
SNS FRAMEWORK/High       [sns_low_lat_client_manager.c    622]
sns_llcm_config - output_buf=0xe04ad000 size=4096 period_us=20000 handle=22
flags=2
FRAMEWORK/Medium     [sns_low_lat_client_manager.c    443]
sns_llcm_find_channel - found first open channel at index 0
SNS FRAMEWORK/Medium     [sns_low_lat_client_manager.c    635]
sns_llcm_config - using channel 0
SNS FRAMEWORK/Medium     [sns_low_lat_client_manager.c    513]
sns_llcm_init_circ_buf_android - max_samples 39
SNS FRAMEWORK/Medium     [sns_stream_service.c    148] Created data stream
to Sensor 'gyro' (b20193b8) from Instance 'llcm' (b256e8a8): b2592d98
SNS FRAMEWORK/Low        [sns_low_lat_client_manager_instance.c   1919]
LLCM - set config complete, rc=0
SNS FRAMEWORK/Medium     [sns_stream_service.c    148] Created data stream
to Sensor 'gyro' (b20193b8) from Instance 'llcm' (b256e8a8): b256a098
SNS FRAMEWORK/Medium     [sns_stream_service.c    148] Created data stream
to Sensor 'resampler' (b201ab68) from Instance 'llcm' (b256e8a8): b255ac68
SNS FRAMEWORK/Medium     [sns_stream_service.c    148] Created data stream
to Sensor 'gyro_cal' (b2016908) from Instance 'llcm' (b256e8a8): b255aff8
```

- Stopping stream

```
SNS FRAMEWORK/Medium     [sns_low_lat_client_manager.c    745]
sns_llcm_disable - stream_id 0
SNS FRAMEWORK/Low        [sns_low_lat_client_manager_instance.c   1919]
LLCM - set config complete, rc=0
```

To collect more LLCM log plackets on QXDM Pro, select the following filtered view:

Log packets – [Message View Configuration] > [Log Packets] > [Known Log Items] > [Common], then click **SNS**.

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9 Dual sensors

To identify dual sensors, three new fields, stream type, HW_ID, and Rigid_Body attributes, have been added in the USTA-TAB. These are displayed on the screen per sensor.

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A References

A.1 Acronyms and terms

Acronym or term	Definition
DRC	Direct report channel
SEE	Sensors Execution Environment
SUID	Sensor unique identifier
USTA	Unified Sensor Test Application

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